

Windows SOC Detection Lab Report

Windows SOC Detection Lab using Wazuh SIEM, Sysmon, and Windows Server 2022

1. Introduction

Security Operations Centers (SOCs) rely on continuous monitoring and analysis of security events to identify suspicious activities and potential threats. This project demonstrates the implementation of a Windows-based SOC Detection Lab using Wazuh SIEM, Sysmon, and Windows Server 2022.

The primary objective of this lab was to collect, analyze, and visualize Windows security events while simulating common administrative and security-related activities. The project provides hands-on experience in log management, event monitoring, threat detection, and security operations workflows.

2. Objectives

The objectives of this project were:

- Deploy a Windows endpoint for monitoring.
- Install and configure Sysmon for enhanced endpoint visibility.
- Connect the endpoint to Wazuh SIEM.
- Collect and analyze Windows security logs.
- Detect user account creation activities.
- Detect administrator group membership changes.
- Monitor privileged logon events.
- Create a security monitoring dashboard for visualization.

3. Lab Architecture

The architecture of the lab consists of the following components:

Windows Server 2022

↓

Sysmon

↓

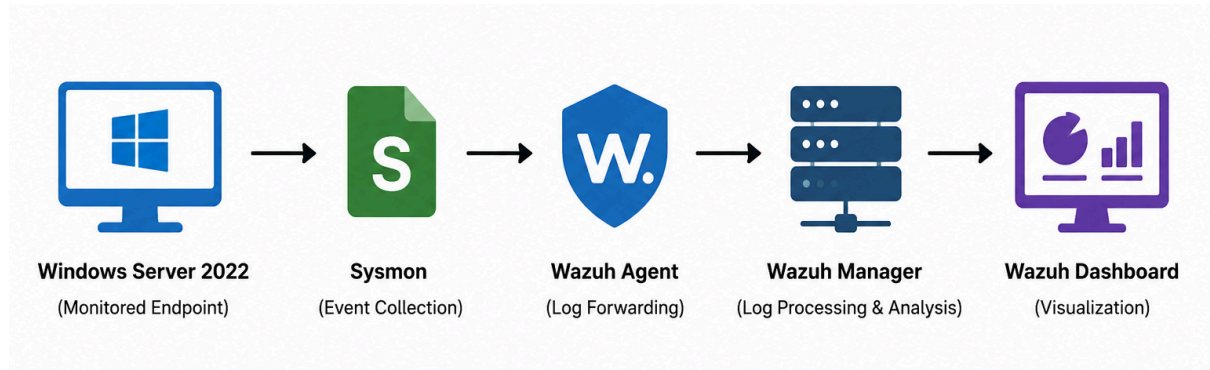
Wazuh Agent

↓

Wazuh Manager

↓
Wazuh Dashboard

Architecture Diagram



4. Technologies Used

Technology	Purpose
Windows Server 2022	Endpoint System
Sysmon	Enhanced Event Logging
Wazuh Agent	Log Collection
Wazuh Manager	Event Processing
Wazuh Dashboard	Visualization and Monitoring
Windows Event Logs	Security Event Source

5. Environment Setup

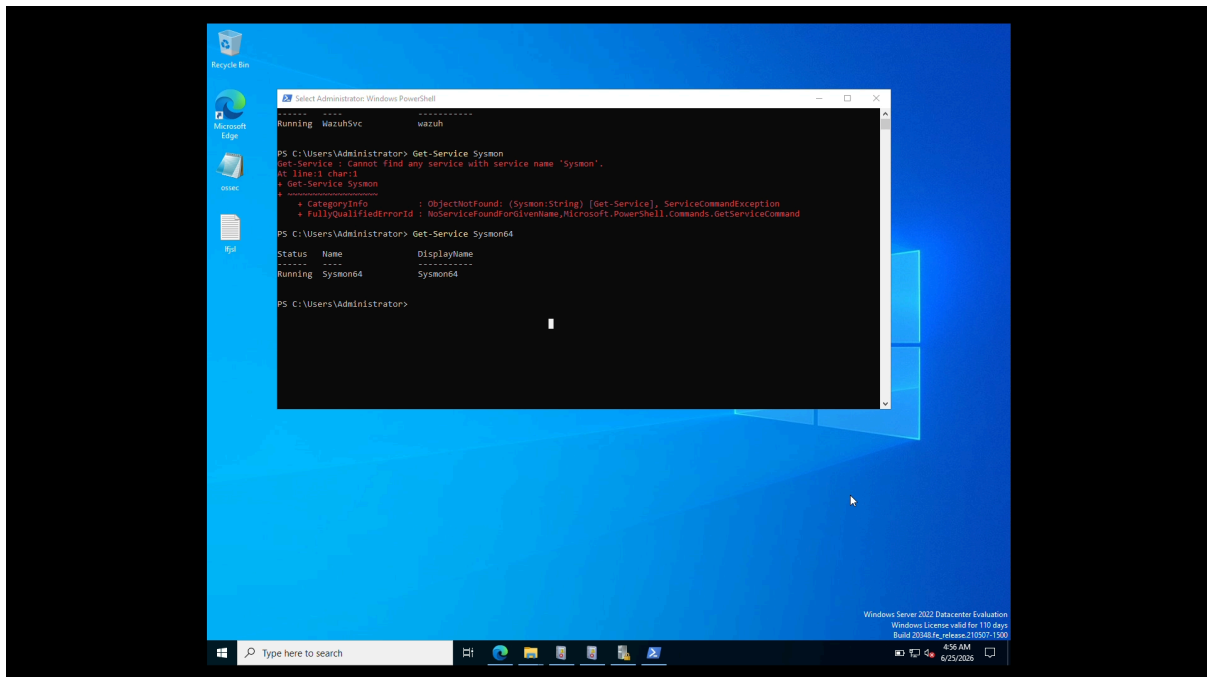
Windows Server 2022

The Windows Server machine was configured as the monitored endpoint.

Sysmon Installation

Sysmon was installed to provide detailed visibility into system activities and process execution events.

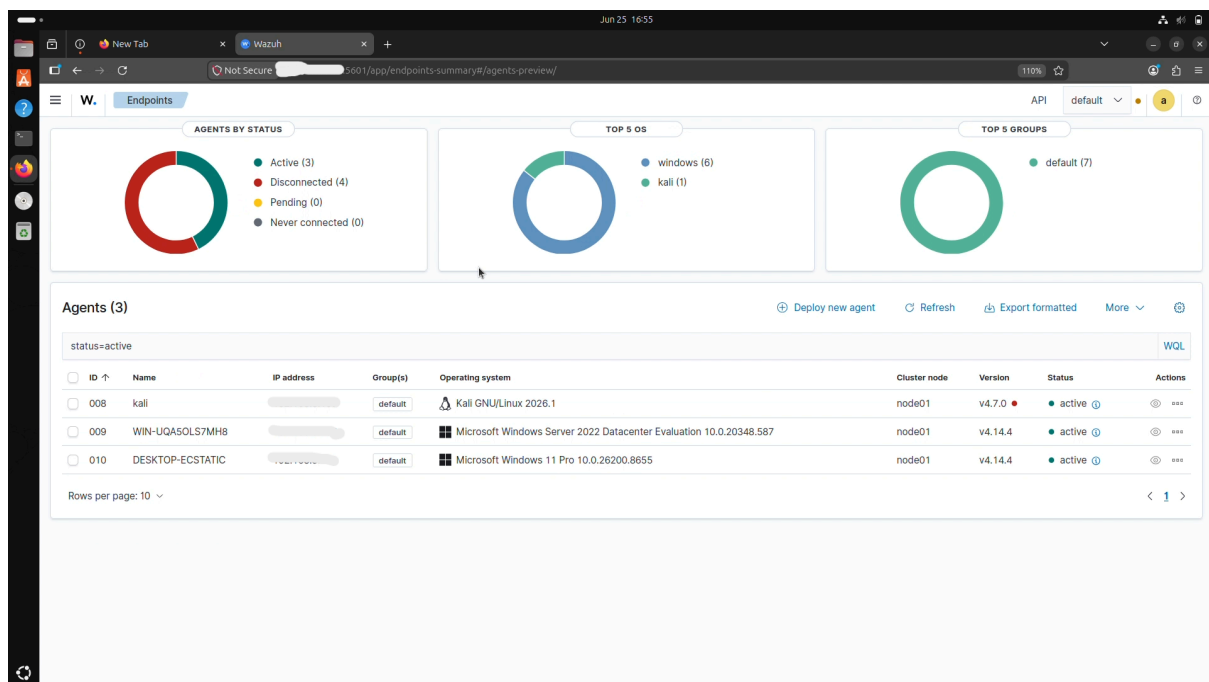
Screenshot



Wazuh Agent Configuration

The Wazuh agent was installed and configured to forward security events to the Wazuh Manager.

Screenshot



6. Detection Scenario 1: Privileged Logon Monitoring

Description

Privileged logon events were monitored to identify administrative account activity within the environment.

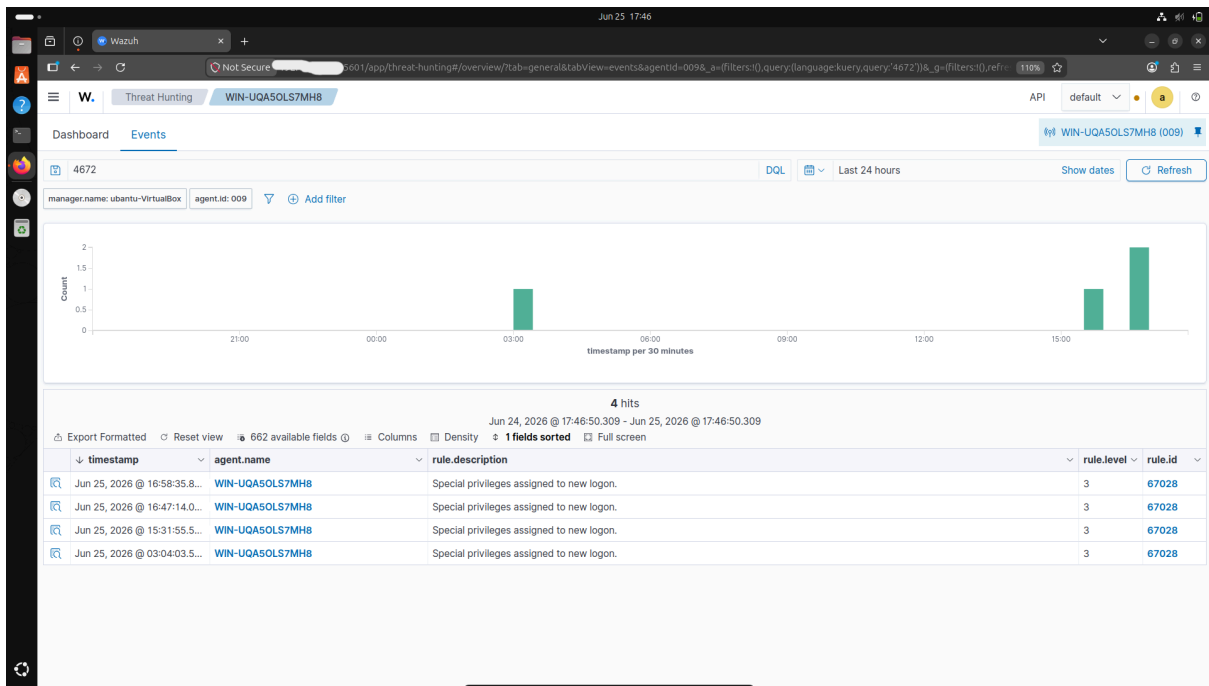
Event ID

4672

Detection Process

- 1. Administrative account login performed.
- 2. Windows generated Event ID 4672.
- 3. Wazuh collected and indexed the event.
- 4. Event was visualized in the dashboard.

Screenshot



7. Detection Scenario 2: User Account Creation

Description

A new user account was created to simulate account provisioning activity.

Event ID

4720

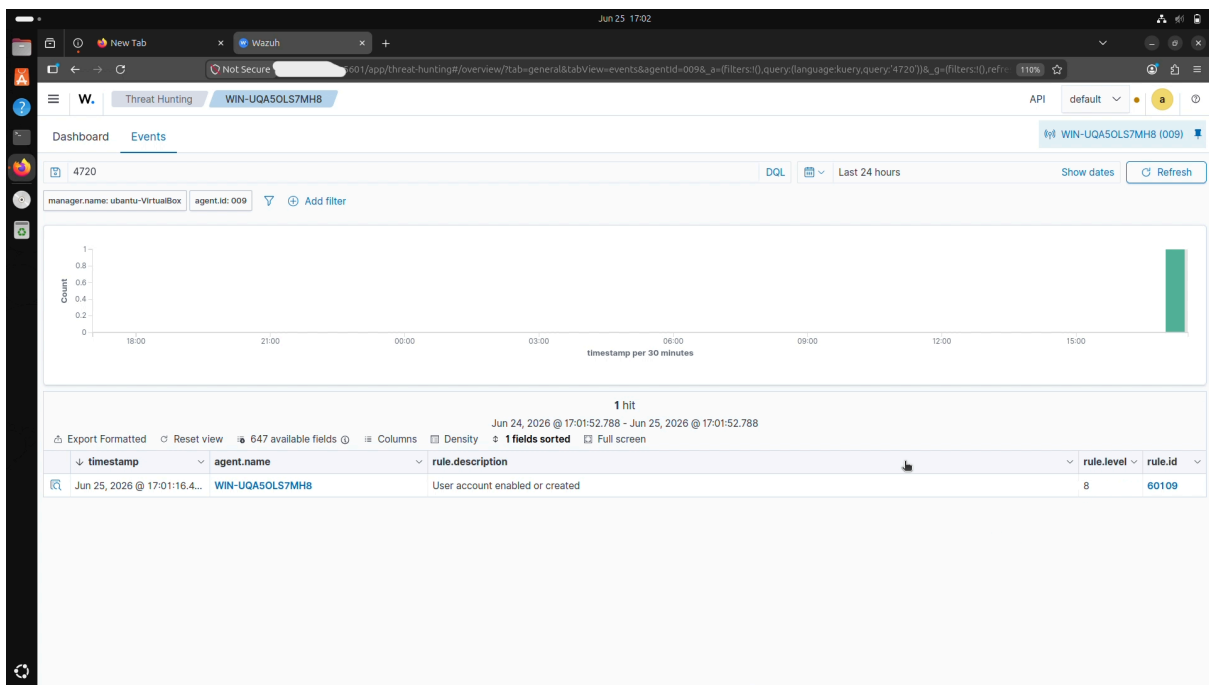
Command Executed

net user hackerxyz P@assw0rd123 /add

Detection Process

- 1. User account created.
- 2. Windows generated Event ID 4720.
- 3. Wazuh detected the event.
- 4. Alert became available for investigation.

Screenshot



8. Detection Scenario 3: Administrator Group Membership Changes

Description

The newly created user was added to the local Administrators group.

Event ID

4732

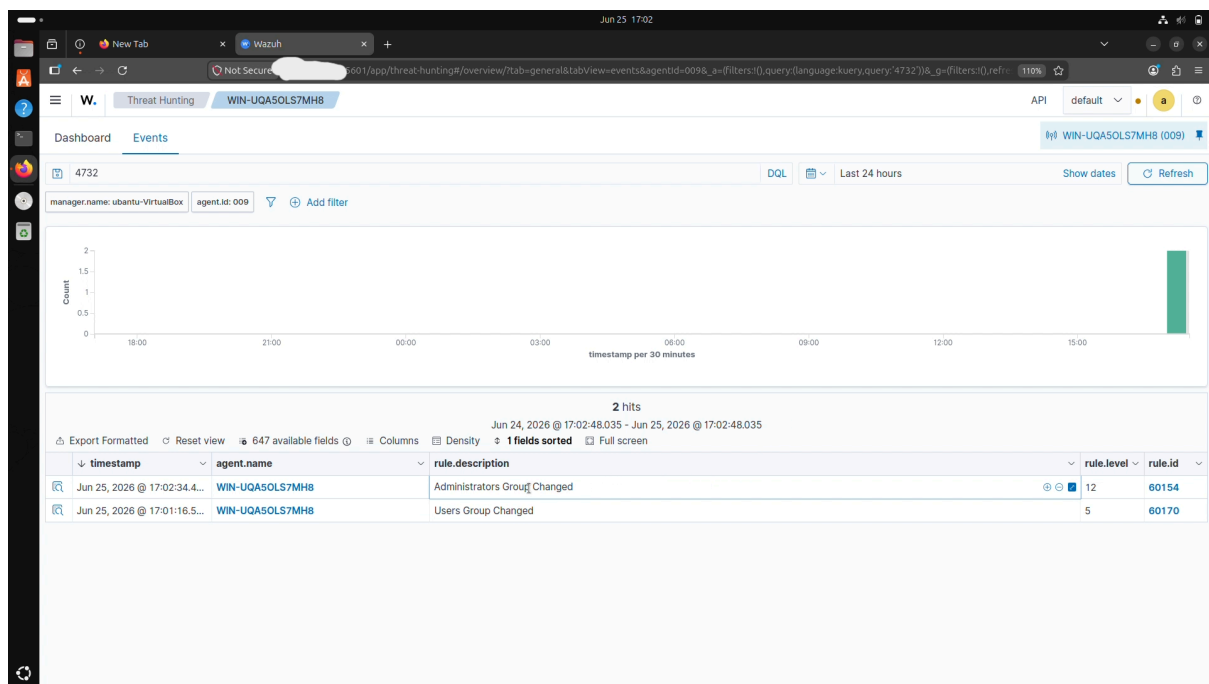
Command Executed

```
net localgroup administrators hacker123 /add
```

Detection Process

1. User added to Administrators group.
2. Windows generated Event ID 4732.
3. Wazuh collected and analyzed the event.
4. Alert was displayed in the dashboard.

Screenshot



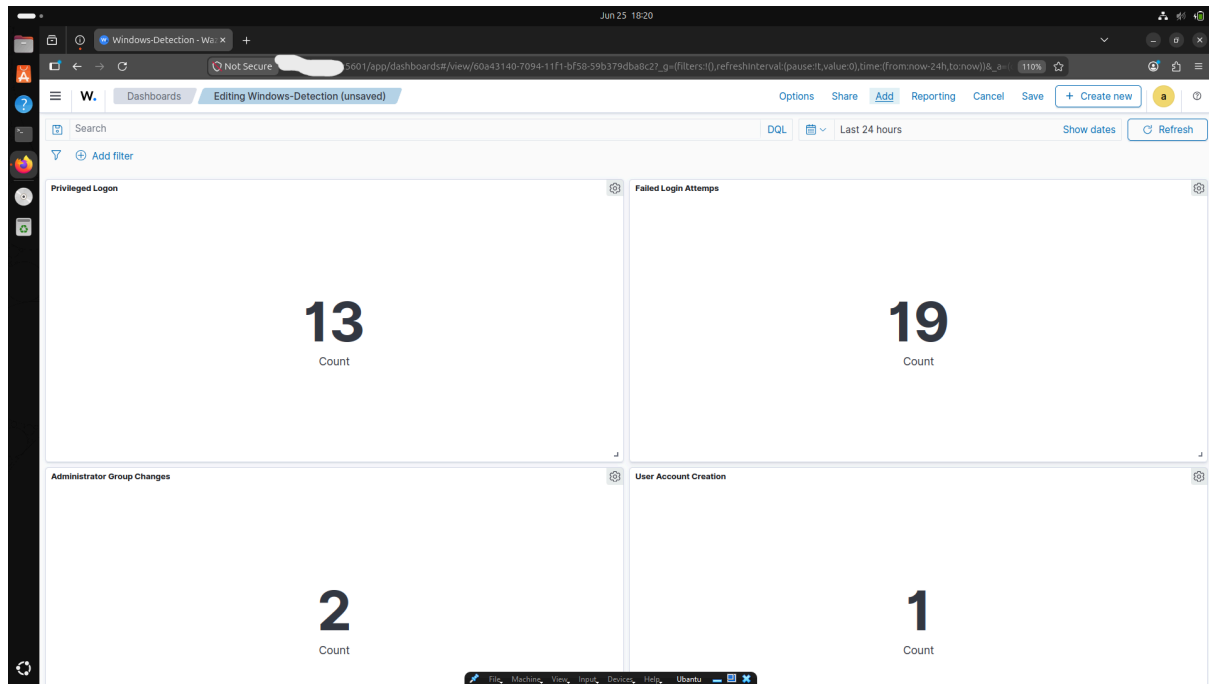
9. Dashboard Visualization

Custom dashboard panels were created to monitor security events.

The dashboard included:

- Privileged Logon Events
- User Account Creation Events
- Administrator Group Changes
- Sysmon Event Monitoring

Dashboard Screenshot



10. Results

The SOC Detection Lab successfully demonstrated:

- Windows Security Event Monitoring
- Endpoint Visibility with Sysmon
- Event Collection using Wazuh Agent
- Security Event Analysis
- Detection of Administrative Activities
- Dashboard Visualization and Monitoring

The lab provided practical experience in SOC operations and security monitoring workflows.

11. Skills Gained

Through this project, the following skills were developed:

- Security Information and Event Management (SIEM)
- Wazuh Administration
- Windows Event Log Analysis
- Endpoint Monitoring
- Security Event Investigation
- Threat Detection
- Blue Team Operations
- SOC Analyst Workflow

12. Conclusion

This project successfully implemented a Windows SOC Detection Lab using Windows Server 2022, Sysmon, and Wazuh SIEM. The environment enabled the collection, analysis, and visualization of security events while demonstrating practical security monitoring capabilities.

The project enhanced understanding of security event monitoring, detection engineering, and SOC operations, providing hands-on experience relevant to modern cybersecurity and Blue Team environments.